YOUYOU WU

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Education

University of Illinois Urbana-Champaign Expected May 2027
B.S. in Mathematics & Computer Science; GPA: 3.91/4.00 *Champaign, IL*

The Chinese University of Hong Kong, Shenzhen September 2023–May 2024
Coursework toward B.Eng. in Computer Engineering; GPA: 3.83/4.00 *Shenzhen, China*

Research Interests

Combinatorics and theoretical computer science. Studying combinatorial objects via algorithmic approaches, using tools like probabilistic methods, graph theory, and randomized and approximation algorithms.

Research Experience

Undergraduate Researcher January 2026–Present
Advised by Deniz Genlik, University of Illinois Urbana-Champaign *Champaign, IL*

Hybrid Multinomials and Their Combinatorics June 2026–Present

- Study hybrid multinomial coefficients arising from intersection products of divisor classes on $\overline{\mathcal{M}}_{0,n}$, including Kapranov degrees.
- Develop recurrences for hybrid products of ψ- and ω-classes and investigate positivity criteria through relaxed Catalan conditions associated with Hessenberg functions.
- Contribute to a collaborative manuscript in preparation.

Border-Strip Decompositions of $3 \times N$ Bracelets and Cylinders January 2026–Present

- Developed a transfer-matrix model for tilings of $3 \times N$ cylinders by oriented length-3 border strips, using 27 boundary states and $\mathrm{tr}(A^N)$ to enumerate taped decompositions.
- Implemented exact enumeration algorithms and analyzed strongly connected components and invariants of the transition digraph; contributed to recurrence formulas and enumeration of untaped bracelets.
- Currently extending the computational framework to higher fixed heights and more general cases.
- Manuscript in preparation.

Undergraduate Researcher, Polymath Jr. 2026 May 2026–Present
Random Plane Graphs; project led by Adam Sheffer *Remote*

- Studied the expected number of leaves in a uniformly random plane spanning tree on arbitrary n-point configuration.
- Jointly proved an $\Omega(\sqrt{n})$ lower bound using a cross-graph charging argument; prepared and checked a complete proof and continue to investigate stronger bounds.
- Collective manuscript in preparation.

Manuscript in Preparation

- Nhi Dinh, Deniz Genlik, Kaustubh Prakash, Maddy Ritter, Eva White, and **Youyou Wu**. *Border-Strip Decompositions of $3 \times N$ Bracelets and Cylinders.*
- add other later!

Advanced Reading & Expository Work

Semi-Random Methods in Combinatorics January 2026–May 2026
Graduate research course with Abhishek Methuku *Champaign, IL*

- Studied recent developments in semi-random methods for hypergraph matchings and related work on expander graphs.
- Prepared and presented a detailed exposition of Gould and Kelly's *Advancing the Rödl Nibble*, including its applications to Latin squares and Steiner systems.

Directed Reading Program in Graph Limit Theory August 2025–December 2025
Mentored by Bowen Li, University of Illinois Urbana-Champaign *Champaign, IL*

- Studied graph homomorphisms, graphons, cut distance, graph convergence, and applications including the Regularity and Removal Lemmas.
- Authored self-contained expository notes in LaTeX and delivered a final presentation on the theory and its principal proofs.
- Delivered a presentation on these materials in undergraduate Math club (SIGma).

Teaching Experience

Course Assistant, Algorithms and Data Structures January 2026–Present
University of Illinois Urbana-Champaign *Champaign, IL*

- Lead weekly lab sections and hold office hours on asymptotic analysis, divide and conquer, graph algorithms, dynamic programming, linear programming, and NP-completeness.
- Develop and deploy homework problems on PrairieLearn and assist students with proofs, algorithm design, and debugging.

Honors & Programs

CS STARS Undergraduate Research Program, University of Illinois Urbana-Champaign Fall 2026
Dean's List, College of Liberal Arts & Sciences Fall 2024, Fall 2025, Spring 2026

Selected Coursework

- **Graduate:** Combinatorics, Semi-Random Methods (Rodl Nibble) and expanders, Mathematical Logic, Real Analysis, Approximation Algorithms, Randomized Algorithms, Computational Complexity
- **Undergraduate:** Computational Geometry, Algorithms, Graph Theory, Honors Abstract Algebra, Honors Real Analysis, Probability and Statistics

Technical Skills

- **Programming:** Python, C++, C, Rust, R **Research tools:** LaTeX, Git, Linux